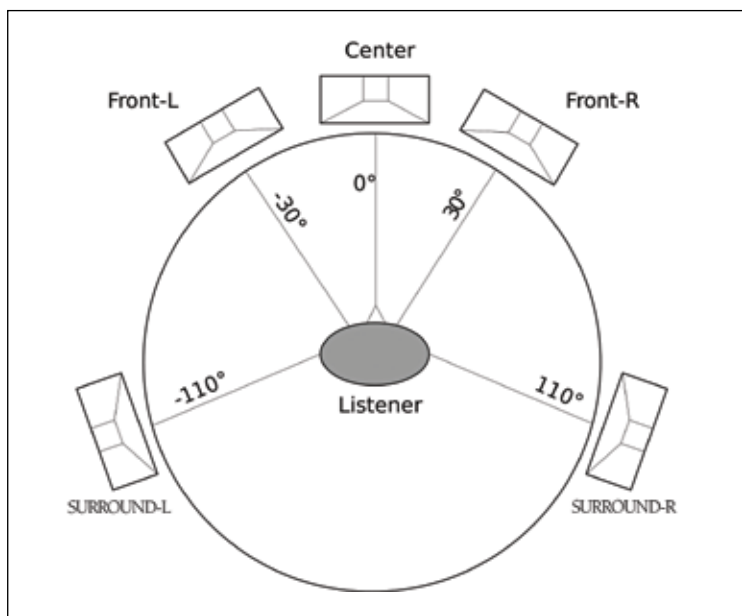




A typical 5.1 surround sound configuration

Effects channel) configuration is the standard for most surround sound applications. This configuration is called the 5.1 system, where the 5 refers to the number of full range channels; including a “.1” to reflect the limited range of the LFE channel. Similarly, any system with x number of full range speakers and y limited range speakers would be denoted by a x.y Surround Sound configuration.

While the full range speaker is used to reproduce sounds across the entire spectrum of audible frequencies, the LFE channel was originally intended to carry extremely low “sub-bass” cinematic sound effects (with commercial subwoofers sometimes going down to 30 Hz, e.g., the loud rumble of thunder or explosions) on their own channel. However, critics argue that this speaker is now a source of some confusion in surround sound, and should not be included in surround sound systems.

The most popular system in use is the 5.1 system, currently employed in many home theatres and a few cinema halls, though most have moved on to a 5.1 system. The most popular audio compression technology, Dolby Digital, encodes data for the 5.1 system, containing 6 channels. These consist